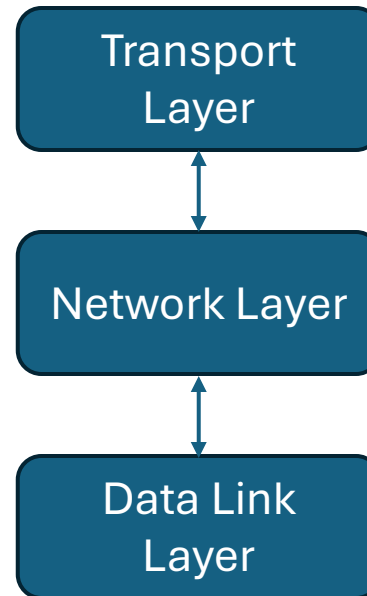


NETWORK LAYER

Module 3

Network Layer:

- The network layer in the TCP/IP protocol suite is responsible for the **host-to-host delivery** of datagrams.
- It **provides services** to the **transport layer** and **receives services** from the **data-link layer**.



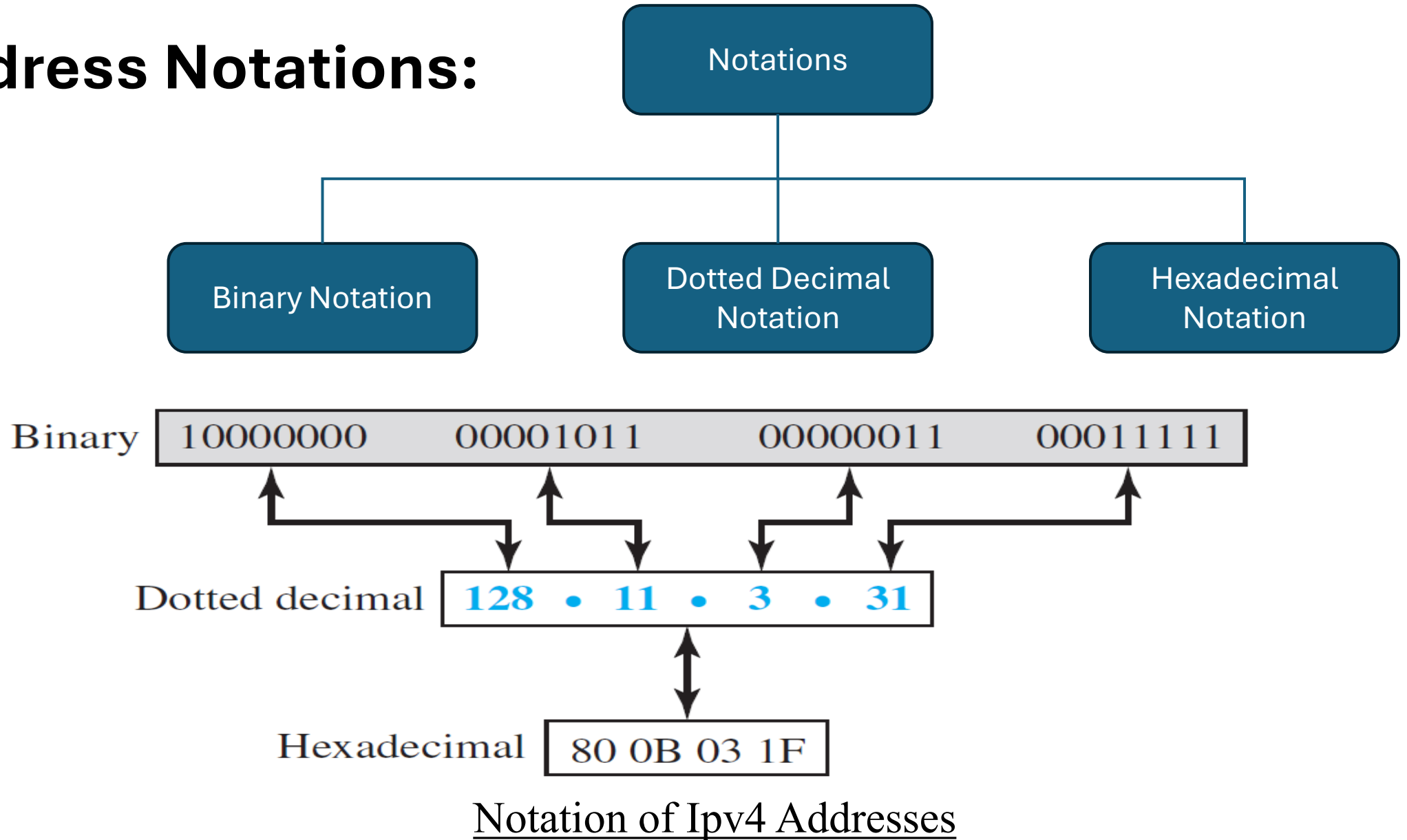
IPv4 Address:

- An IP address is a **unique address** used to identify a **device's connection** to the **Internet**.
- An IPv4 address is a **32-bit address** that **uniquely identifies** the **connection** of a **host** or **router** to a **network**.
- It **identifies the connection**, not the **device**, so it **may change** when the **device moves** to **another network**.
- A **device** with **multiple network connections** can have **multiple IP addresses**.

Address Space:

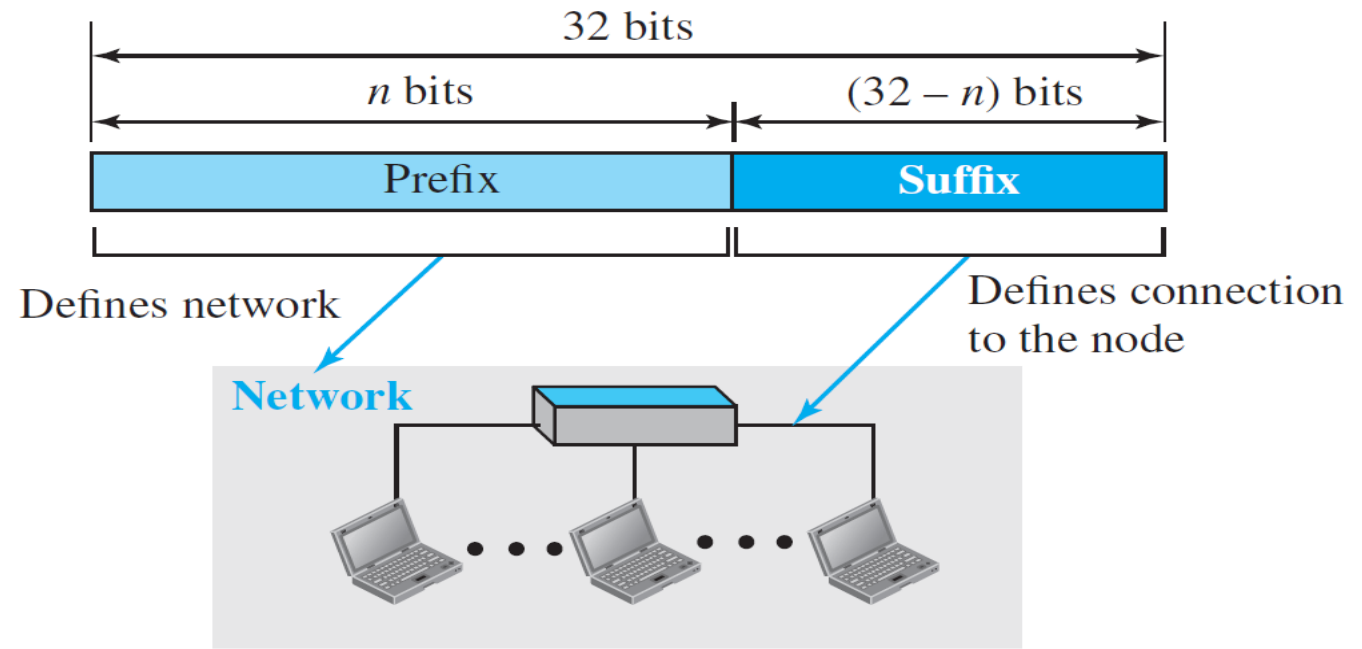
- An address space is the **total number of addresses** available in a protocol. If an address uses **b bits**, the address space is 2^b , since each bit can be **0 or 1**.
- IPv4 uses **32-bit addresses**, giving $2^{32} = 4,294,967,296$ (over 4 billion) **possible addresses**. Thus, without any restrictions, **IPv4** could **provide addresses for more than 4 billion devices**.

IPv4 Address Notations:

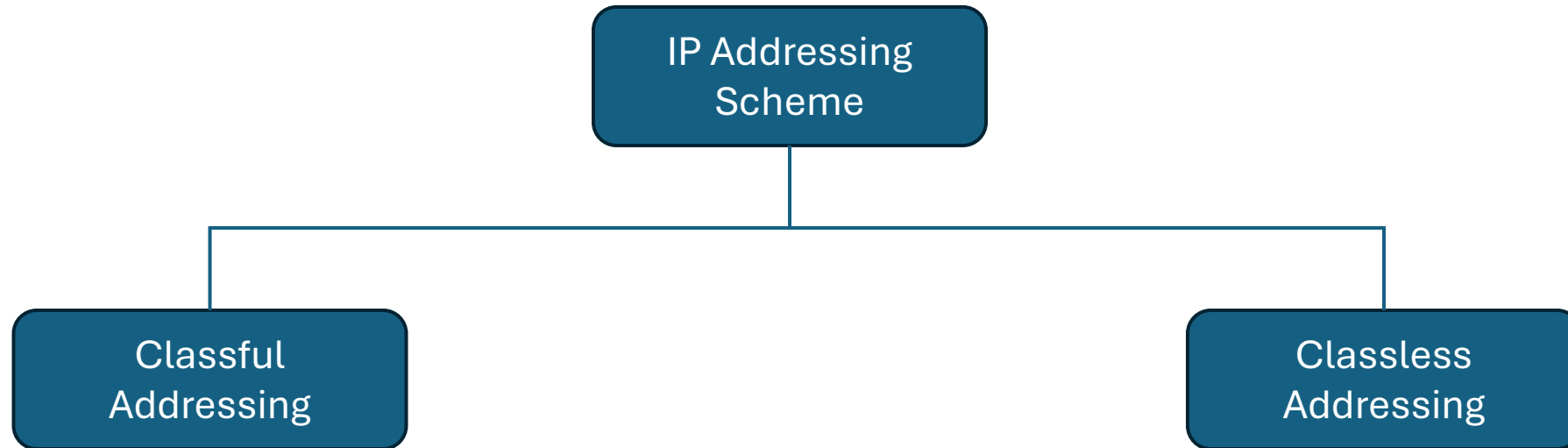


Hierarchy in Addressing:

- A 32-bit IPv4 address is hierarchical, but divided only into two parts. The first part of the address, called the prefix, defines the network; the second part of the address, called the suffix, defines the node (connection of a device to the Internet).
- Figure shows the prefix and suffix of a 32-bit IPv4 address. The prefix length is n bits and the suffix length is $(32 - n)$ bits.

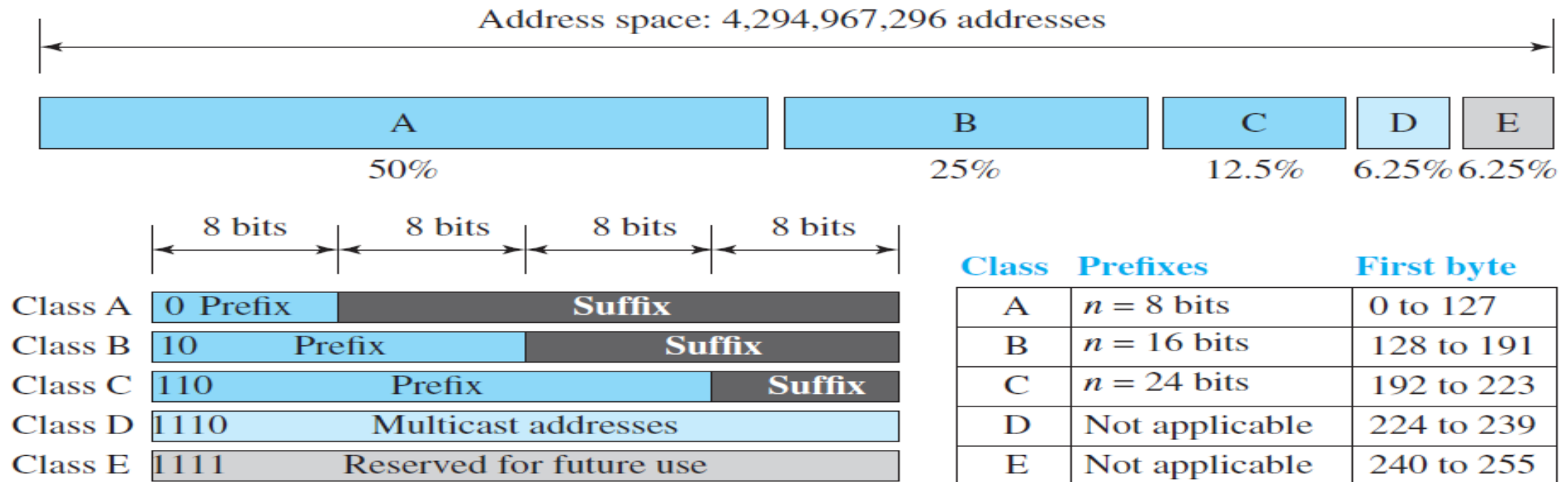


IP addressing schemes:



Classful Addressing:

- The whole **IPv4 address space** is divided into **five classes** (class A, B, C, D, and E). This scheme is referred to as **classful addressing**.



- Prefix** means **network bits** and **suffix** means **host bits**.

- **Class A:** The network portion is 8 bits, but the first bit 0 identifies the class, leaving 7 bits for the network identifier. Thus, there are $2^7 = 128$ possible networks. Network numbers 0 and 127 are reserved for special purposes. The remaining 24 bits are for hosts, giving $2^{24} - 2 = 16,777,214$ hosts per network.
- **Class B:** The network portion is 16 bits, but the first two bits 10 identify the class, leaving 14 bits for the network identifier. Thus, there are $2^{14} = 16,384$ networks. The remaining 16 bits are for hosts, giving $2^{16} - 2 = 65,534$ hosts per network.
- **Class C:** The network portion is 24 bits, but the first three bits 110 identify the class, leaving 21 bits for the network identifier. Thus, there are $2^{21} = 2,097,152$ networks. The remaining 8 bits are for hosts, giving $2^8 - 2 = 254$ hosts per network.

- **Class D:** Addresses beginning with 1110 belong to Class D. They are used for multicast and are not divided into network and host portions.
- **Class E:** Addresses beginning with 1111 belong to Class E. They are reserved for experimental/future use and are not divided into network and host portions.

Example: If the IP address of the system is 64.123.46.125, to which class it belongs to?

Solution: Class A

Network Mask or Default Mask:

- It is a mathematical tool which is used to solve the networking problem. In the representation of network mask, **network bits** will be **1** and **hosts bits** will be **0**.

For **Class A**: **255.0.0.0**

For **Class B**: **255.255.0.0**

For **Class C**: **255.255.255.0**

- Performing **bit-wise AND** with **IP address** and **mask** give **network ID**.

Example: If the IP address of the system is 64.123.46.125, what will be the network ID and directed broadcast address?

Solution: Since 64.123.46.125 belongs to the class A, then the network mask for class A is 255.0.0.0.

Network ID = 64.123.46.125 (bitwise AND) 255.0.0.0 = 64.0.0.0 (to **get netID**, set **all host bits** as **0**)

DBA = 64.255.255.255 (to get the **DBA**, set **all host bits** as **1**)

Drawbacks of Classful Addressing:

- Class A can be assigned to only 128 organizations in the world, but each organization needs to have a single network (seen by the rest of the world) with 16,777,216 nodes (computers in this single network). Since there may be only a few organizations that are this large, most of the addresses in this class were wasted (unused).
- Class B addresses were designed for midsize organizations, but many of the addresses in this class also remained unused.
- Class C addresses have a completely different flaw in design. The number of addresses that can be used in each network (256) was so small that most companies were not comfortable using a block in this address class.
- Class E addresses were almost never used, wasting the whole class.

Address Depletion Problem:

- Address depletion means the **rapid shortage** of **IPv4 addresses** due to **inefficient classful addressing**.
 - **Class A:** **Very large blocks** (~16.7 million addresses) were given to only a **few organizations**, causing **many addresses** to be **wasted**.
 - **Class B:** **Medium-sized blocks** were also often **underused**.
 - **Class C:** Blocks were **too small** (256 addresses), so many organizations **did not find** them **suitable**.
 - **Class E:** Was **reserved** and **rarely used**, **wasting** a **large address space**.
- Therefore, Classful addressing became **outdated** because it **wasted** many **IP addresses** and **caused** a **shortage** of **IPv4 addresses**.

Subnets:

- Dividing a **network** into **smaller networks** for **efficient utilization** of the **IP addresses** is known as **subnetting**.
- During subnetting, **subnet bits** are **borrowed** from the **host** portions.
- Performing **bitwise AND** of the **IP address** and **subnet mask**, we get the **subnet ID**.

Problem 1: In a Class C network, if subnet mask is 255.255.255.240. Find the number of subnets and number of hosts in each subnets.

Solution 1:

For a Class C network:

Default Class C subnet mask = 255.255.255.0

Given subnet mask = 255.255.255.240

Step 1: Convert the last octet to binary

240 in binary: $240 = 11110000$

So the subnet mask is: 255.255.255.11110000

Compared with the default Class C mask:

255.255.255.00000000

We borrowed 4 bits for subnetting.

Step 2: Number of subnets

Number of subnets = $2^4 = 16$

Step 3: Number of hosts per subnet

There are: $8-4=4$ bits left for hosts.

Hosts per subnet = $2^4 - 2 = 16 - 2 = 14$ hosts

Problem 2: In a Class B network, if subnet mask is 255.255.248.0. Find the number of subnets and number of hosts in each subnets.

Problem 2: In a Class B network, if subnet mask is 255.255.248.0. Find the number of subnets and number of hosts in each subnets.

Number of subnets = $2^5 = 32$, Number of hosts/subnet = $2^{11} - 2 = 2046$ hosts/subnet

References:

Data Communications AND Networking by BEHROUZ A. FOROUZAN.



THANK YOU